Missing-verb illusion in Turkish center-embeddings? An investigation of case interference and phrase lengths

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Introduction

- This study investigates doubly center-embedded structures with relative clauses (2-CE-RCs) in Turkish.
  - *E.g., The rat that the cat that the dog chased ate died.* (Chomsky & Miller, 1963, p. 286)
- 2-CE-RCs are reported to be extremely difficult to process due to (among many other factors)
  - working memory limitations (Gibson,1998; 2000)
  - similarity-based interference (Lewis & Vasishth, 2005)
  - non-optimal prosodic phrase lengths (Fodor, 2013)
- Due to their extreme processing difficulty, the omission of the second verb in 2-CE-RCs can be unnoticed, resulting in a grammaticality illusion:
  - *The patient who the nurse who the clinic had hired ~~admitted~~ met Jack.* (Frazier, 1985, p. 178)
- Missing-verb illusion has been confirmed in verb-initial languages;
  - English (e.g., Christiansen & MacDonald, 2009; Gibson & Thomas, 1999)
  - French (Gimenes et al., 2009)
- Findings on verb-final languages are inconclusive;
  - they either support its presence, e.g., German (Bader, 2015; Häussler & Bader, 2015)
  - or not, e.g., German and Dutch (Frank & Ernst, 2018, Frank et al., 2016; Vasishth et al., 2010)
- In verb-final languages more robust predictions for upcoming verbs may prevent the illusion.

The Present Study

- Turkish, a highly inflectional and head-final language, can provide further data on similarity-based interference, prosodic phrase lengths and missing-verb illusion.

- The present study investigates:

  - if the processing difficulty of Turkish center-embeddings can be facilitated by similarity-based interference and/or prosodic phrase lengths,

  - if there is a missing-verb illusion in Turkish center-embeddings,

  - if the existence of missing-verb illusion could be modulated by similarity-based interference and/or prosodic phrase lengths.

- Two acceptability judgment tasks were conducted:

  - Expt. 1: tests similarity-based interference and missing-verb illusion, 80 participants

  - Expt. 2: tests prosodic phrase lengths and missing-verb illusion, 38 participants

Expts. 1 & 2: Materials & Procedure

- Experimental sentences were Turkish 2-CE-RCs nested as a complement clause inside a matrix clause:

Ø	[Marangoz-lar-ın	[nakliyeci-ler-in	[kiracı-nın	gri	koltuğ-u	yerleştir-diğ-i	oda-ya]
Pro	carpenter-PL-GEN	mover-PL-GEN	tenant-GEN	gray	sofa-ACC	place-PART-3SG	room-DAT
	NP1	NP2	NP3			VP3	

taşı-dık-ları	dolab-ı]	kur-duk-ları-nı]	bil-iyor-um.
move-PART-3PL	cabinet-ACC	set.up-VN-3PL-ACC	know-PROG-1SG
VP2		VP1	

'*I know that the carpenters set up the cabinet that the movers moved to the room that the tenant placed the gray sofa in.*'

- Expt. 1: four conditions manipulating
  - grammaticality (grammatical vs. ungrammatical due to a missing VP2 (in blue above))
  - syntactic interference (high vs. low interference manipulated with case marking on the subject NPs)
- Expt. 2: four conditions manipulating
  - grammaticality as in Expt. 1
  - prosodic phrase lengths (encouraging vs. discouraging phrase lengths for the optimal phrasing of the sentence)
- Comprehension questions about different subject-verb relations followed the sentences.
- 24 experimental sentences & 36 fillers in each experiment
- Task: rate the sentences on a scale of 1 to 5 (5, acceptable)

Results

	Expt. 1			Expt. 2		
	LowInt	HighInt	Accuracy	ENC	DISC	Accuracy
Grammatical	2.59 (.05)	2.86 (.05)	78%	2.87 (.09)	2.69 (.09)	81%
Ungrammatical	2.40 (.08)	2.48 (.08)	65%	2.63 (.08)	2.54 (.08)	66%

Cumulative link mixed effects models with R package *ordinal* (Christensen, 2018) showed:

Expt. 1: HighInt > LowInt (β = .40, *SE* = .13, *z* = 3.04)

Grammatical > Ungrammatical (β = .36, *SE* = .17, *z* = 2.15)

HighInt: Grammatical > Ungrammatical (β = .54, *SE* = .22, *z* = 2.45)

LowInt: Grammatical = Ungrammatical (β = .14, *SE* = .22, *z* = .67)

Expt. 2: ENC > DISC (β = .25, *SE* = .14, *z* = 1.76)

Grammatical > Ungrammatical (β = .36, *SE* = .19, *z* = 1.88)

ENC: Grammatical > Ungrammatical (β = .48, *SE* = .25, *z* = 1.92)

DISC: Grammatical = Ungrammatical (β = .21, *SE* = .20, *z* = 1.03)

Conclusions

- **Similarity-based interference:**
  - Syntactic interference affected processing Turkish center-embeddings but in the unpredicted direction. Why?
    - judgment (i.e., offline) data may reveal limited effects of case interference (e.g., Avetisyan et al., 2020)
    - the non-factive reading in the LowInt conditions may have caused an extra processing difficulty (Özyıldız, 2017)
  - An eye-tracking experiment with sentences excluding non-factive interpretation verbs showed the predicted pattern (data in the Q&A session).

- **Prosodic phrase lengths:**
  - Optimal and relatively balanced phrase lengths eased the processing of Turkish center-embeddings (Fodor, 2013)

- **Missing-verb illusion:**
  - Overall no missing-verb illusion in Turkish 2-CE-RCs; verb-final languages may be less susceptible to the illusion (Frank & Ernst, 2018; Frank et al., 2016; Vasishth et al., 2010)
  - But the perceived difficulty of the structure had an effect; the illusion may indeed be cross-linguistic (Gibson & Thomas, 1999), appearing in the relatively more difficult constructions (e.g., DISC conditions in Expt. 2)

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